

08 — Backend Architecture & Service Layer Reference

■ Backend Structure & Module Overview

The backend of the Maritime Port Intelligence Platform is built on **FastAPI** (**CODE_PH_0**) and running asynchronously on **Uvicorn** (**CODE_PH_1**).

```
backend/ ■■■ app/ ■ ■■■ main.py # FastAPI Application Entry Point & Router Assembly ■ ■■■ routes/ #
HTTP Controller Router Layer ■ ■ ■■■ cargo.py # Cargo Forecasting & Scenario Simulation Routes ■ ■
■■■ vessels.py # AIS Vessel Tracking & Congestion Routes ■ ■ ■■■ trade.py # Trade Lanes & Commodity
Prices Routes ■ ■ ■■■ anomaly.py # Anomaly Detection Event Routes ■ ■ ■■■ incentive.py # Trade
Incentive & Monte Carlo Simulation Routes ■ ■ ■■■ twin.py # Digital Twin Scenario Routes ■ ■ ■■■
copilot.py # AI Copilot Query & Dispatcher Routes ■ ■ ■■■ pipeline.py # Ingestion Pipeline & Log Routes
■ ■ ■■■ executive.py # Executive KPI & Revenue Trend Routes ■ ■ ■■■ security.py # SOC Threats, DPDP
Compliance & Audit Logs Routes ■ ■ ■■■ upload.py # Custom CSV File Upload Controller ■ ■ ■■■
forecast.py # Intelligence Forecasting Router ■ ■ ■■■ health.py # Health Check Route ■ ■■■ services/
# Analytical Business Logic & Analytical Engines ■ ■■■ forecasting.py # Holt-Winters + Ridge Regression
ML Cargo Engine ■ ■■■ synthetic_data.py # Port Metric Generators & Monte Carlo Engine ■ ■■■ copilot.py
# LangGraph Cognitive 3-Tier Query Dispatcher ■ ■■■ nmpa_vessels.py # AIS Tracking & Vessel Route Risk
Engine ■ ■■■ security.py # MITRE ATT&CK & Cryptographic Audit Logger
```

■ Detailed Breakdown of Every Backend File

File: **CODE_PH_0**

- **Purpose:** Application entry point. Instantiates **CODE_PH_0**, configures CORS middleware, mounts all 12 feature routers, and defines root status endpoint.
- **Input:** Server startup invocation (**CODE_PH_0**).
- **Processing:** Registers router prefixes (**CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**, **CODE_PH_3**, etc.).
- **Output:** Running HTTP ASGI server instance on port 8000.
- **Used by:** Uvicorn server launcher & Streamlit HTTP frontend.

File: **CODE_PH_0**

- **Purpose:** Exposes REST endpoints for cargo volume forecasting, commodity metadata, backtesting model evaluation, forecast explainability drivers, and What-If scenario simulations.
 - **Input:** HTTP GET/POST queries (**CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**, scenario JSON payloads).
 - **Processing:** Validates input parameters via FastAPI **CODE_PH_0** and **CODE_PH_1** helpers and delegates computations to **CODE_PH_2**.
 - **Output:** JSON REST responses containing forecast values, prediction intervals, WAPE metrics, and simulation results.
 - **Used by:** Streamlit Tab 3 (**CODE_PH_0**).
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File: *CODE_PH_0*

- **Purpose:** Serves live AIS vessel tracking data, ETA countdowns, and berth congestion alert logs.
 - **Input:** HTTP GET request.
 - **Processing:** Calls *CODE_PH_0*.
 - **Output:** JSON list of 25 active vessels with coordinate locations (*CODE_PH_0*, *CODE_PH_1*), delay probabilities, and route risk scores.
 - **Used by:** Streamlit Tab 1 & Tab 2 (*CODE_PH_0* & *CODE_PH_1*).
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File: *CODE_PH_0*

- **Purpose:** Exposes endpoints for commodity trade lane growth, Bloomberg commodity price indices, and AI-ranked market opportunities.
 - **Input:** HTTP GET request.
 - **Processing:** Delegates data extraction to *CODE_PH_0*.
 - **Output:** JSON payload of trade routes, growth YoY %, price trends, and revenue potentials.
 - **Used by:** Streamlit Tab 4 (*CODE_PH_0*).
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File: *CODE_PH_0*

- **Purpose:** Serves active anomaly detection logs and historical severity frequency distributions.
 - **Input:** HTTP GET request.
 - **Processing:** Calls *CODE_PH_0*.
 - **Output:** JSON list of classified anomaly events (Surge, Decline, Congestion, Weather) with severity ratings (*CODE_PH_0*).
 - **Used by:** Streamlit Tab 5 (*CODE_PH_0*).
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File: *CODE_PH_0*

- **Purpose:** Serves RL trade incentive recommendations and accepts POST requests for 1,000-iteration Monte Carlo revenue simulations.
 - **Input:** HTTP GET and POST JSON payloads (*CODE_PH_0*, *CODE_PH_1*, *CODE_PH_2*).
 - **Processing:** Calls Monte Carlo probabilistic engine in *CODE_PH_0*.
 - **Output:** P10, P50, P90 revenue quantiles and sample arrays for Plotly histogram.
 - **Used by:** Streamlit Tab 6 (*CODE_PH_0*).
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File: *CODE_PH_0*

- **Purpose:** Serves virtual port digital twin scenario stress-test results.

- **Input:** HTTP GET scenario key (**CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**, **CODE_PH_3**).
 - **Processing:** Computes congestion index, berth/storage utilization, workforce deltas, and radar comparison points.
 - **Output:** JSON dictionary of baseline vs scenario metrics.
 - **Used by:** Streamlit Tab 7 (**CODE_PH_0**).
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File: **CODE_PH_0**

- **Purpose:** Handles conversational queries directed to the AI Maritime Copilot.
 - **Input:** HTTP POST query payload (**CODE_PH_0**).
 - **Processing:** Dispatches query through **CODE_PH_0**.
 - **Output:** JSON payload containing 3-tier dispatch trace, confidence score, and answer text.
 - **Used by:** Streamlit Tab 8 (**CODE_PH_0**).
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File: **CODE_PH_0**

- **Purpose:** Accepts multipart form uploads for custom CSV reports.
 - **Input:** Multipart file binary stream (**CODE_PH_0**).
 - **Processing:** Validates **CODE_PH_0** extension, checks size limit (max 20MB), saves file to **CODE_PH_1** directory, and validates CSV structure using Pandas.
 - **Output:** Success status JSON with file path, row count, and parsed column headers.
 - **Used by:** External API clients & data integration tools.
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File: **CODE_PH_0** (666 lines)

- **Purpose:** Core analytical ML engine of the platform. Loads **CODE_PH_0**, performs time-series feature engineering, fits Holt-Winters Exponential Smoothing + Ridge Regression models, computes 95% dynamic prediction interval bounds, evaluates WAPE/MAPE/MAE/RMSE out-of-sample backtesting metrics, maps NMPA berths, and executes scenario simulations.
 - **Input:** Filter parameters (**CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**) or scenario vectors.
 - **Processing:** Advanced Pandas time-series resampling, NumPy matrix operations, Scipy/Holt-Winters math.
 - **Output:** Structured Python dictionary consumed directly by **CODE_PH_0**.
 - **Used by:** **CODE_PH_0** & **CODE_PH_1**.
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File: **CODE_PH_0** (720 lines)

- **Purpose:** Generates realistic simulation data for berth statuses, trade lane momentum, commodity price indices, anomaly events, Monte Carlo revenue distributions, executive KPIs, and digital twin scenarios.
- **Input:** Function parameters and random seeds.
- **Processing:** NumPy statistical sampling, Gaussian noise distributions, deterministic pseudo-random seed controls.
- **Output:** Python dictionaries for trade, anomaly, executive, twin, and incentive endpoints.

- Used by: `CODE_PH_0`, `CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`, `CODE_PH_4`.

File: `CODE_PH_0` (600 lines)

- **Purpose:** Implements the LangGraph 3-tier cognitive query dispatcher.
- **Tier 1 (Heavy Synthesis LLM):** Complex multi-variable queries.
- **Tier 2 (Fast Reasoning SLM):** Domain specific cargo queries.
- **Tier 3 (Deterministic Zero-LLM):** Fallback rule engine for instant responses.
- **Input:** User string query.
- **Processing:** Pattern matching, intent classification, simulated step-by-step dispatch trace generation, agent assignment.
- **Output:** Answer text, confidence score, data references, and trace steps.
- **Used by:** `CODE_PH_0`.

■ Request-Response Lifecycle

```
1. Client Sends Request ■■■ HTTP GET /cargo/forecast?horizon=6&commodity=TOTAL%20COAL 2. FastAPI
Middleware ■■■ CORS validation & URL path matching 3. Router Dispatcher ■■■ Calls
`routes/cargo.py:get_forecast()` 4. Business Service ■■■ Invokes
`forecasting.py:get_enhanced_cargo_forecast()` 5. Data Extraction ■■■ Reads
`data/port_cargo_monthly.csv` & filters `TOTAL COAL` 6. ML Execution ■■■ Fits Holt-Winters + Ridge
model, calculates 95% bounds & WAPE 7. Response Structuring ■■■ Pydantic/FastAPI serializes dictionary
to JSON 8. HTTP Response Sent ■■■ Returns HTTP 200 OK + JSON payload to Client
```